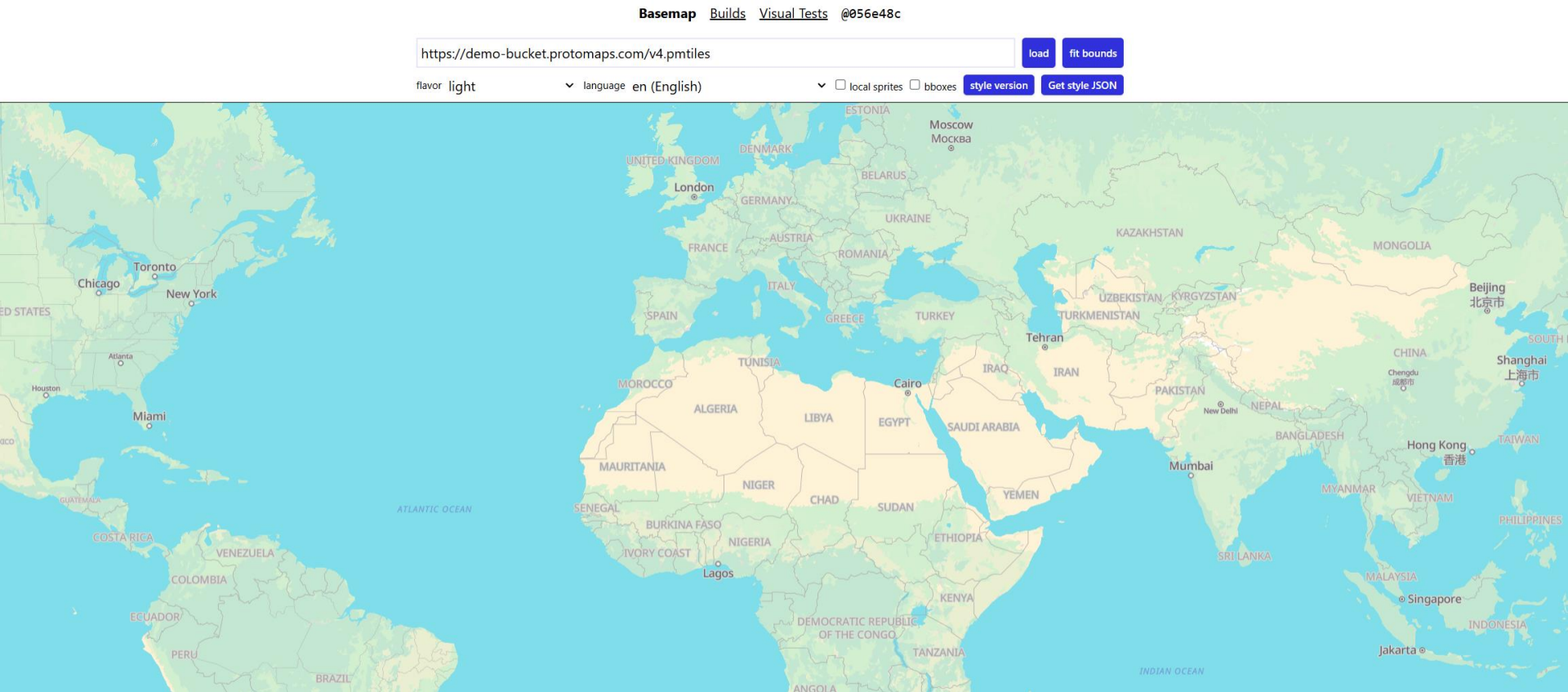


Open Source Mapping & Raster Visualisation



Protomaps global OpenStreetMap demo, <https://maps.protomaps.com>

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Overview

- 1. Open Source Alternatives to Mapbox**
- 2. Hosting Large Spatial Data Files Without Cloud Services**
- 3. Mapping Raster Data Online**

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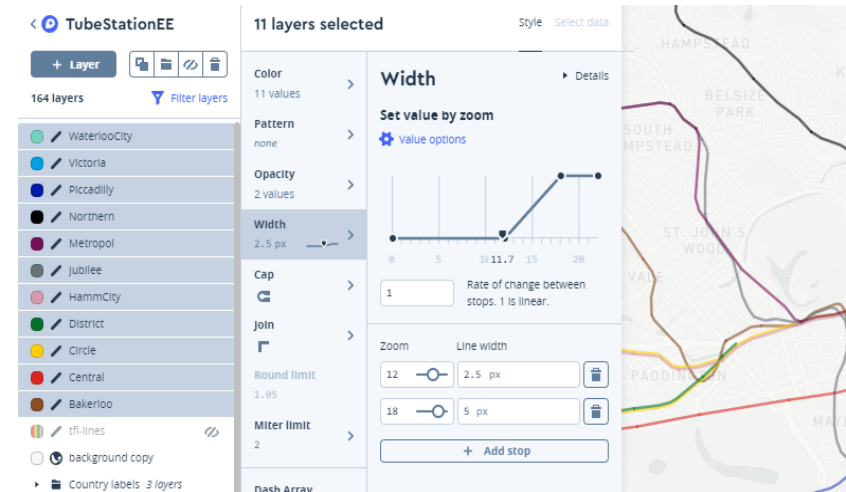
Mapbox is a Great Tool, But Not Open Source

Best Cloud Service for Cartography

Can quickly create high quality map visualisations and basemaps using Mapbox Studio. Includes default OpenStreetMap layers, and uses Tilesets to host your own data as vector tiles.

Popular Tool with Good Functionality and Online Documentation

We have used Mapbox to create lots of different types of interactive maps and visualisations. Useful tool with lots of options and examples.



MapBox Studio provides sophisticated cartographic tools for custom basemap design.

Mapbox has Limitations for Research

Mapbox is a commercial company. You can create an account for free, but there is a cap on total visitor numbers on the free subscription tier. The rules for the free tier may change in the future. You can transfer your Mapbox Studio designs to other platforms (export Style JSON files) but some limitations.

Mapbox No Longer Open Source

Mapbox began as an open source project, but switched to a proprietary license in 2020. You are supposed to only use Mapbox software alongside a working Mapbox account.

Some Projects Need to be Fully Open Source

Some government and research projects require the use of fully open source tools. Mapbox no longer fulfils this requirement.

MapLibre Open Source Alternative

When Mapbox switched to a commercial license, an open source alternative launched called MapLibre, forked from the last open source version of Mapbox JS.

MapLibre is Very Similar to Mapbox, Easy to Change

You don't need to learn a new library, as the code is almost exactly the same as the examples we have been working on. Just need to switch a few tags to MapLibre tags-

<https://maplibre.org/maplibre-gl-js/docs/guides/mapbox-migration-guide/>

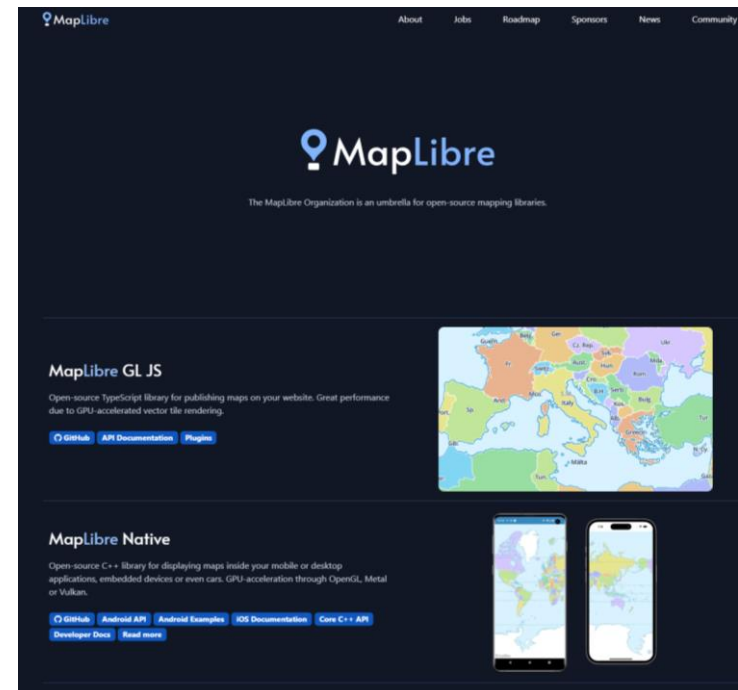
Good Online Documentation and Examples

Popularity of MapLibre has grown over time and there is now good examples and documentation at-

<https://maplibre.org/maplibre-gl-js/docs/examples/>

But No Vector Tiles Hosting Service or Mapbox Studio

The main challenge with switching to MapLibre is that we lose the Mapbox Studio tool and the vector tile hosting (Tileset) services that Mapbox provides. Need another hosting approach...



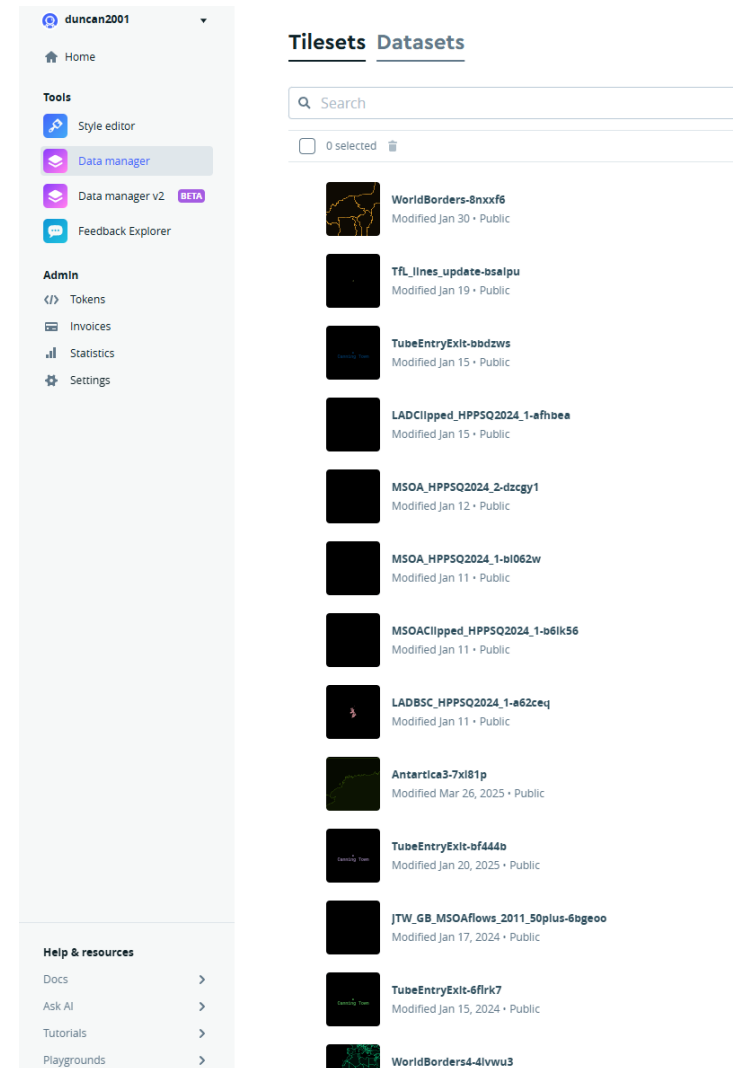
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Using Mapbox to Create & Host Vector Tiles

Mapbox has a very useful spatial data hosting service that provides a scalable solution for large spatial data files as vector tiles.

If we don't use Mapbox then we need an alternative hosting approach.



Remember GeoJSON Only Works for Small Files

Small Spatial Datasets: GeoJSON

Simple web maps can use text-based vector formats like GeoJSON, stored on standard web space (cheap and easy). However GEOJSON does not scale for large volumes of data.

Spatial Data Scaling

Mapping the entire globe at multiple zooms levels is a lot of data! Vector tiles are a scalable solution to this problem, dividing the world into a grid at multiple zoom levels, efficiently serving spatial data at the required spatial scale to users.

Standard Options: Self-Hosting or Cloud Service

Self-Hosted Route

Data is stored and served on your own server. Most powerful approach: can have as much data as you want, custom cartography, total control.

Typically requires spatial database to store the data, and serve the data to the client. Technically more demanding approach. Open Source tools available (PostGIS, Mapnik, PMTiles...).

Cloud Mapping Services Route

Your map data is stored and served by a cloud service provider. You still need to define the cartography and to develop the HTML/JavaScript web interface. But server programming skills not required.

Leading providers: MapBox, Carto, MapTiler, ArcGIS Online. This module teaches MapBox, and some Leaflet.

New Alternatives: PMTiles & MLTiles

What if we had a flat file format that was scalable like a database?

Protomap Tiles: PMTiles

General format for tiled data addressed by Z/X/Y coordinates. Can be vector tiles or raster data. Uses HTTP Range Requests to fetch only the relevant tiles inside a PMTiles archive, so scalable. Designed for efficient panning and zooming, compatible with MapLibre and Leaflet- <https://docs.protomaps.com/pmtiles/>

PMTiles is a read-only format. It is not possible to update an archive in-place without re-writing the entire file: does not have transactional functionality of a true database.

MapLibre Tiles: MLTiles

New format released in 2026, similar to PMTiles. Not widely supported yet, but one to watch- <https://github.com/maplibre/maplibre-tile-spec>

PMTiles Performance: Testing OpenStreetMap

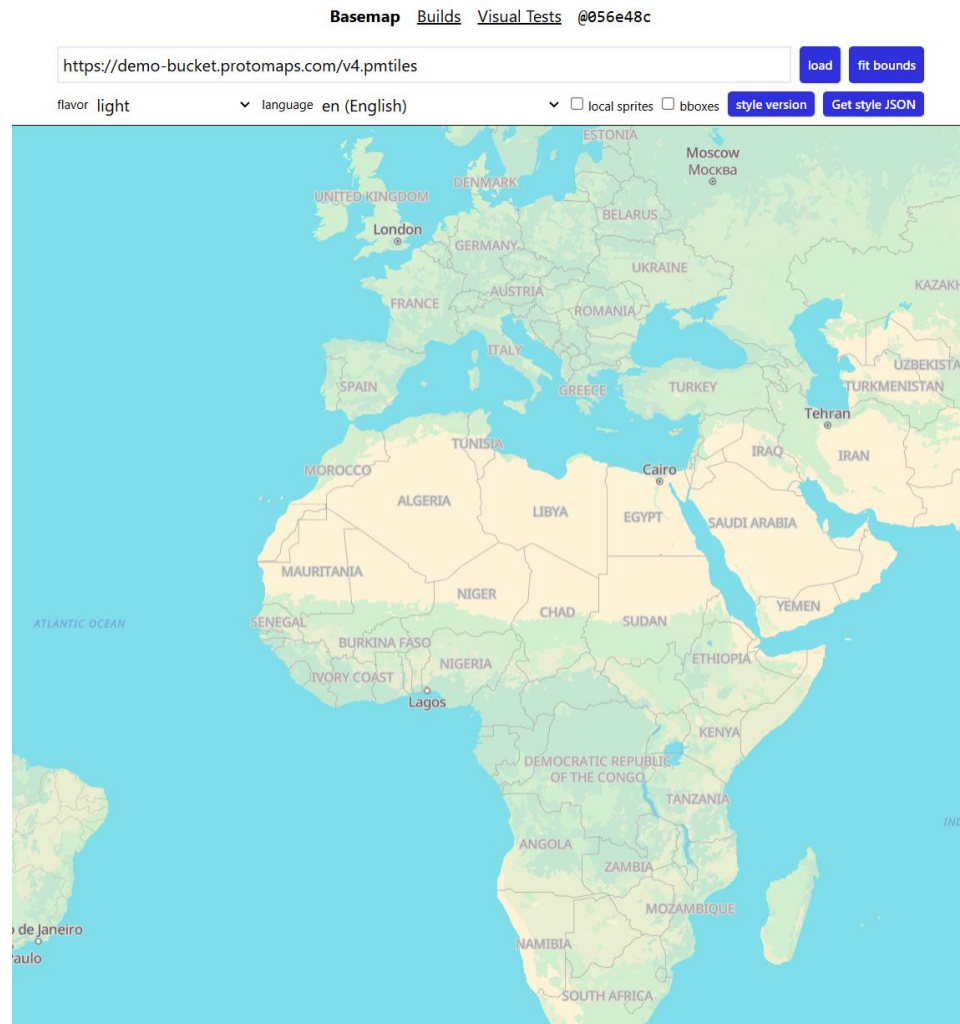
Protomaps website has OSM vector tiles for the entire globe-

<https://maps.protomaps.com/>

This is 125 gigabytes of data served as vector tiles from a PMTiles file in the browser. Performance is impressive.

As the data uses vector tiles, the cartography can be easily changed using style JSON files-

<https://docs.protomaps.com/basemaps/flavors>



Testing PMTiles: Ordnance Survey ZoomStack

Tested on standard web hosting for Ordnance Survey ZoomStack data (2.5gb)-
<https://luminocity3d.org/CASATest/MapLibrePMTiles3OSZoomStackOutdoors.html>

Again the performance is good. [Ordnance Survey ZoomStack](#) is an open dataset of vector tiles of OS data with zoom level styling for web mapping. Available through Digimap as MBTiles file, then converted to PMTiles.

To visualise PMTiles, you need to use MapLibre or Leaflet JS as client library, with PMTiles plugin-
<https://docs.protomaps.com/pmtiles/maplibre>



How to Create PMTiles

PMTiles Command Line Tool

Protomaps provide a command line tool for data conversion-

<https://docs.protomaps.com/pmtiles/cli>

This will convert MBTiles files to PMTiles. It will also allow you to extract OpenStreetMap data as PMTiles from the latest global OSM PMTiles build file. It does not generalise spatial data for different zoom levels.

Create PMTiles from GeoJSON: Tippecanoe

This is a flexible tool that can create PMTiles files from many types of spatial data such as GeoJSON and CSV files. It is very good at scaling vector data to be visible at multiple zoom levels. MacOS tool (can also run via Docker)-

<https://github.com/felt/tippecanoe>

Tippecanoe created by Erica Fischer who previously worked for Mapbox. You can also create MBTiles files using QGIS, but generally Tippecanoe is more reliable (QGIS good for raster tiles creation).

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Raster Data Visualisation Important for Many Tasks

We have focussed on vector data, but raster data is relevant for many environmental research projects using remotely sensed data and imagery.

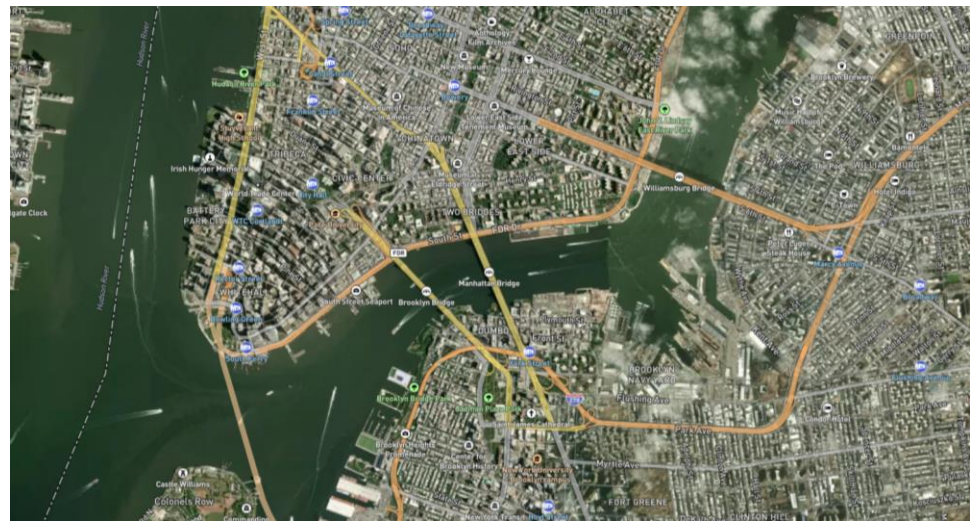
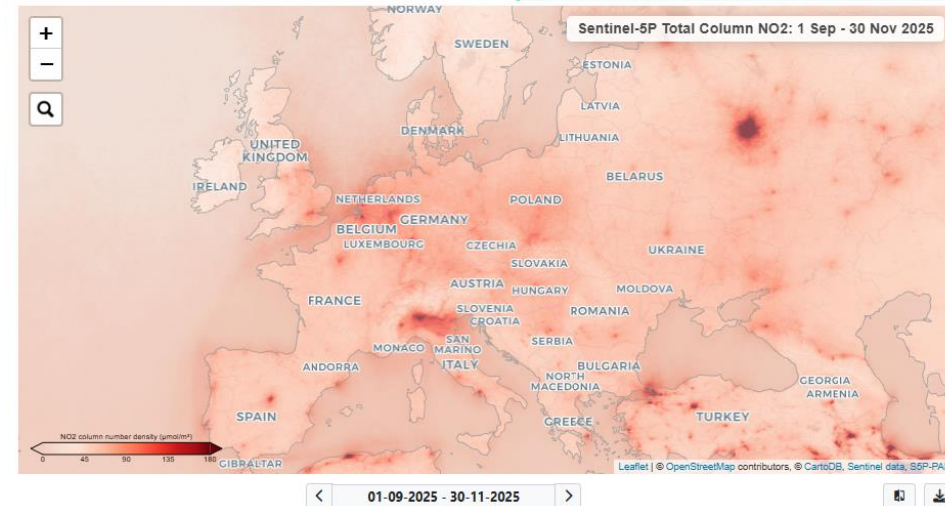
Global raster datasets can get very large (100s of gigabytes), and similar to vector data, we need to consider appropriate hosting solutions.

Top right: ESA Sentinel NO2 Data Map-
<https://maps.s5p-pal.com/no2/>

Bottom Right: Mapbox Satellite Streets-
<https://www.mapbox.com/maps/satellite>

Sentinel-5P Total Column NO2

This map is based on **OPERATIONAL** Sentinel-5P L2 products.



Hosting Options for Raster Webmapping

Cloud Services & APIs

Major providers of remotely sensed data serve tiles via APIs, e.g. Google Earth Engine, NASA, Mapbox. Some flexibility limitations. Mapbox will host raster data, uploaded in GeoTIFF format. Straightforward solution, but Mapbox is not open source.

Flat File XYZ Directory

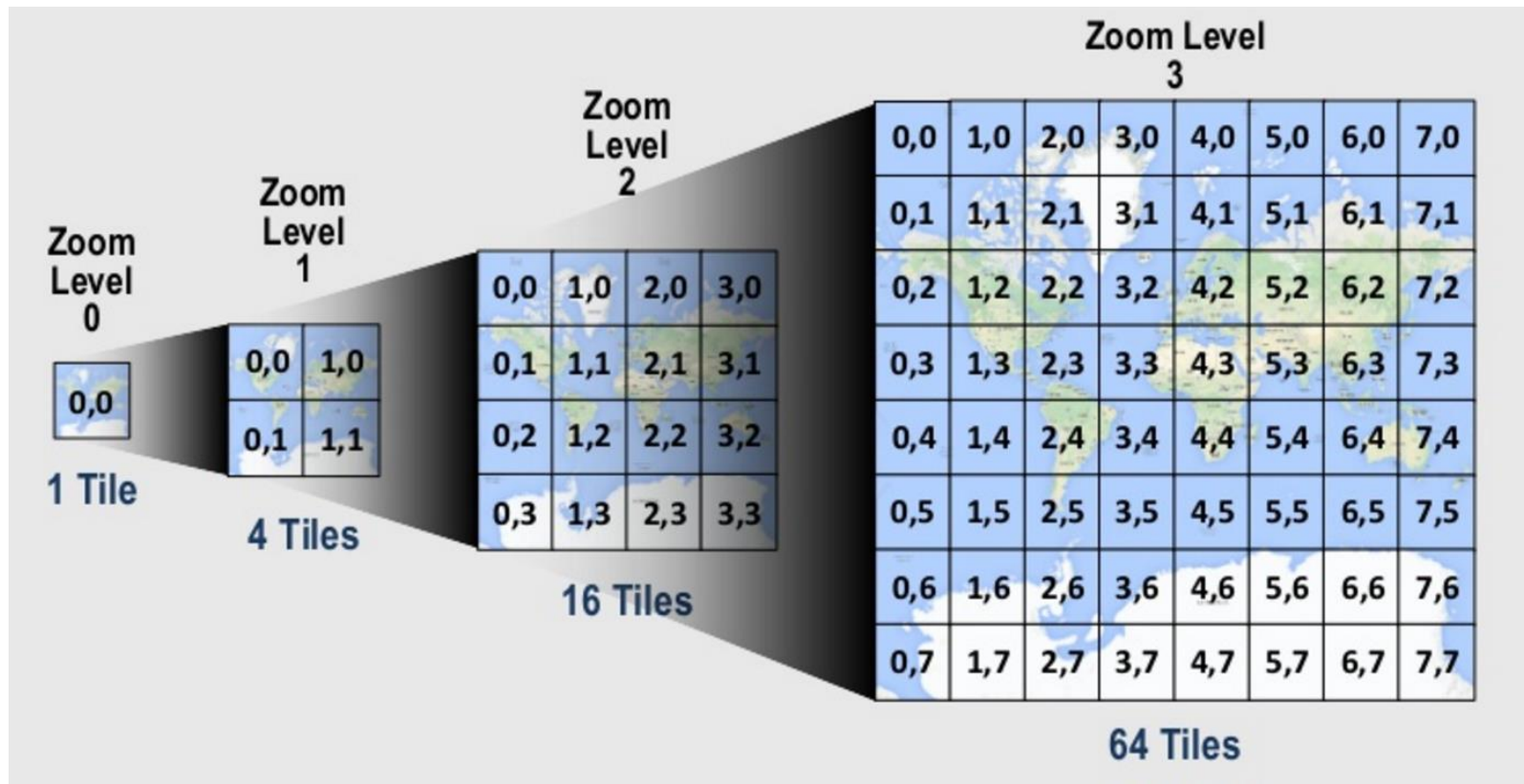
Store files directly on standard webhosting. XYZ data can be generated in QGIS and R/Python libraries. At high zoom levels, millions of files that can exceed hosting limits.

Scalable Flat File Formats: PMTiles & Cloud GeoTIFF

Vector data solution also works for rasters. Single file and higher compression than XYZ hosting.

Revision: Raster Map Tiles

Regular grid of georeferenced images at multiple zoom levels (quadtree).
Standard scalable format established by Google Maps using fixed Web Mercator projection. Pre-rendered small (256 or 512) images quick to serve. Provide the specific location and zoom level requested by the client.



File Sizes for Tiled Global Raster Data

Test of Global Human Settlement Layer Global Raster data at different zoom levels.
Global raster data becomes very large beyond zoom level 8, with millions of tiles-

Scale	Zoom Level	Number Tiles (global)	Total XYZ Size (Gb)	Total PMTiles Size (Gb)
Global	0	1	0	0
	1	4	0	0
	2	16	0	0
	3	64	0.01	0.01
Continental	4	256	0.02	0.02
	5	1,024	0.06	0.05
National	6	4,096	0.19	0.16
	7	16,384	0.62	0.48
Regional	8	65,536	1.94	1.34
	9	262,144	6.13	3.55
City	10	1,048,576	19.50	9.50
	11	4,194,304	62.00	25.40

<https://luminocity3d.org/WorldPopDen/>

Conclusions

Mapbox Excellent Tool but Has Limitations

Free tier has restrictions, and Mapbox is no longer an open source project. Some projects require fully open source tools.

MapLibre is a Great Alternative

Forked from Mapbox so very easy to switch. But also need an alternative for hosting large spatial datasets.

Protomap Tiles Scalable Flat File Solution

Good performance and works with MapLibre and Leaflet. Good solution for open source projects.

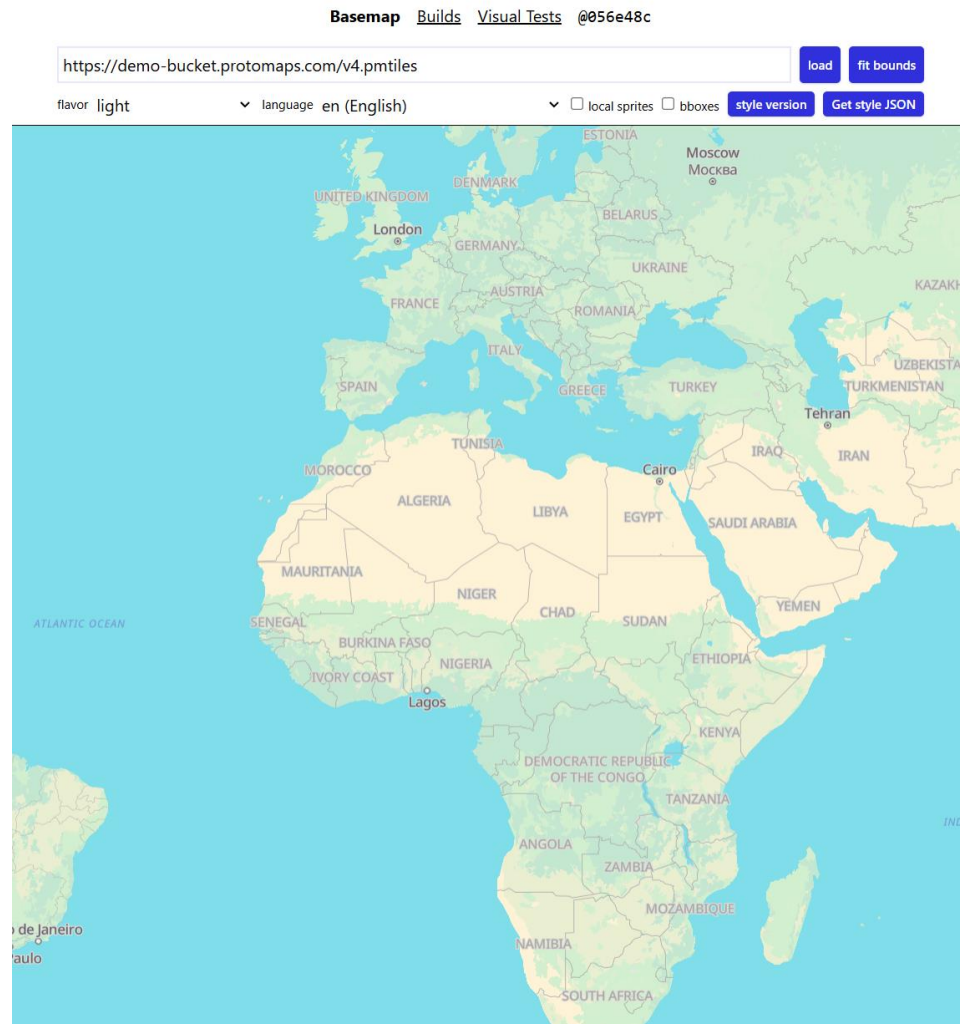
Raster Data Important for Some Projects

Environmental data and imagery raster based. Again creates hosting challenges. Can use cloud services, or PMTiles also works for raster data.

Practical: PMTiles with MapLibre and Leaflet

In the practical we will use MapLibre with PMTiles to create an OpenStreetMap global map without cloud hosting. And we will change the style of this basemap.

Then we will use Leaflet and PMTiles to create a global raster map.



Continue to Work on Group Projects

Continue to work on developing your group project ideas, and identifying datasets to use and visualisations to create.

Each group needs to post on Slack with their project concept idea and their group members.

Any students who haven't confirmed their group, contact duncan.a.smith@ucl.ac.uk or speak in person now.